

# (1)

## Honors Physics 632 WI 11 HW6 Solutions

$$1) |\psi(0)\rangle = \frac{1}{\sqrt{2}} (|100\rangle - |111\rangle) ; r \leq a, E_{nl} = \frac{\hbar^2 k_{ln}^2}{2m}$$

$$(2) P(E_{nl}) = \sum_{m=-l}^l |\langle nlm | \psi(t) \rangle|^2$$

$$= \frac{1}{2} \delta_{nl} (\delta_{l0} + \delta_{l1})$$

$$P(E_{10}) = \frac{1}{2}, \quad P(E_{11}) = \frac{1}{2}, \quad \text{all others zero}$$

$$(b) |\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle = \frac{1}{\sqrt{2}} \left( e^{-iE_{100}t/\hbar} |100\rangle - e^{-iE_{111}t/\hbar} |111\rangle \right)$$

$$\langle \vec{r} | \psi(t) \rangle = \frac{1}{\sqrt{2}} \left( e^{-iE_{100}t/\hbar} \phi_{100}(\vec{r}) - e^{-iE_{111}t/\hbar} \phi_{111}(\vec{r}) \right)$$

$$\phi_{100}(\vec{r}) = A_{10} j_0(k_{10}r) Y_0^0(\theta, \phi)$$

$$\phi_{111}(\vec{r}) = A_{11} j_1(k_{11}r) Y_1^1(\theta, \phi)$$

$$(c) \langle \hat{R} \rangle = \int_{r \leq a} d^3r \, r |\psi(\vec{r}, t)|^2$$

In the cross terms we encounter  $Y_0^{0*} Y_1^1$  etc.  
These lead to:

$$\int_0^{2\pi} d\phi \, e^{i\phi} = 0 \Rightarrow \text{No cross terms survive.}$$

$$\text{Also } \int d\Omega Y_l^{m*} Y_{l'}^{m'} = \delta_{ll'} \delta_{mm'}$$

(c) cont.  $\langle \hat{R} \rangle = \frac{1}{2} \int_0^a r^3 dr \left[ |A_{10}|^2 (j_0(k_{10}r))^2 + |A_{11}|^2 (j_1(k_{11}r))^2 \right]$

- need to evaluate further only if given tables.

(d) Prob that  $z > 0$  :  $P_{z>0} = \int_0^a r^2 dr \int_0^{\pi/2} \sin \theta d\theta \int_0^{2\pi} d\phi | \psi(\vec{r}, t) |^2$

- cross terms do not survive because  $\int d\Omega y_0 y_1 = 0 \dots$

$$P_{z>0} = \frac{2\pi}{2} \int_0^a r^2 dr \int_0^{\pi/2} \sin \theta d\theta \left\{ |A_{10}|^2 (j_0(k_{10}r))^2 \left[ \frac{1}{4\pi} \right] + |A_{11}|^2 (j_1(k_{11}r))^2 \left[ \frac{1}{4} \left( \frac{3}{2\pi} \right) \sin^2 \theta \right] \right\}$$

Use  $\int_0^a r^2 dr |A_{nl}|^2 (j_l(k_{nl}r))^2 = 1$  and

$$\int_0^{\pi/2} \sin \theta d\theta = \int_0^1 dx = 1 ; \int_0^{\pi/2} \sin \theta d\theta \sin^2 \theta = \int_0^1 dx (1-x^2) = \frac{2}{3}$$

$$P_{z>0} = \frac{2\pi}{2} \left\{ 1 \times \frac{1}{4\pi} + \frac{2}{3} \times \left[ \frac{1}{4} \left( \frac{3}{2\pi} \right) \right] \right\}$$

$$P_{z>0} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

2) hydrogen;  $| \psi(0) \rangle = \frac{1}{\sqrt{2}} | 100 \rangle + \frac{1}{\sqrt{3}} | 210 \rangle + \frac{1}{\sqrt{6}} | 21-1 \rangle$

(a)  $| \psi(t) \rangle = e^{-iHt} | \psi(0) \rangle$ ;  $\hbar = 1$

~~$| \psi(t) \rangle = \frac{1}{\sqrt{2}} e^{-iE_1 t} | 100 \rangle + \frac{1}{\sqrt{3}} e^{-iE_2 t} | 210 \rangle + \frac{1}{\sqrt{6}} e^{-iE_2 t} | 21-1 \rangle$~~

(b)  $P_{z>0} = \int d^3r | \psi(\vec{r}, t) |^2 \Theta(z)$

Again we still have  $\int_0^{2\pi} d\phi \rightarrow m=m'$  in any surviving cross-terms.  $\Rightarrow$  only  $| 100 \rangle + | 210 \rangle$

With  $m=m'$ ,  $\int_0^{2\pi} d\phi e^{i(m-m')\phi} = \int_0^{2\pi} d\phi = 2\pi$

$$P(z>0) = 2\pi \int_0^\infty r^2 dr \int_0^{\pi/2} \sin\theta d\theta \left\{ \frac{1}{2} R_{10}^2(r) \left( \frac{1}{4\pi} \right) + \right.$$

$$\frac{1}{3} R_{21}^2(r) \left( \frac{1}{4} \cdot \frac{3}{\pi} \cos^2\theta \right) + \frac{1}{6} R_{21}^2(r) \left( \frac{1}{4} \cdot \frac{3}{2\pi} \sin^2\theta \right)$$

$$\left. + \frac{1}{\sqrt{6}} R_{10}(r) R_{21}(r) \left( \frac{1}{\sqrt{4\pi}} \cdot \frac{1}{2} \sqrt{\frac{3}{\pi}} \cos\theta \right) \left( e^{-i(E_2-E_1)t} + e^{-i(E_2-E_1)t} \right) \right\}$$

Use:  $\int_0^\infty r^2 dr R_{n\ell}^2(r) = 1$ ,  $\int_0^1 dx x^2 = \frac{1}{3}$ ,  $\int_0^1 dx (1-x^2) = \frac{2}{3}$

$$\int_0^1 dx x = \frac{1}{2}$$

$$P(z > 0) = 2\pi \cdot \frac{1}{2} \cdot \frac{1}{4\pi} \cdot 1 + 2\pi \cdot \frac{1}{3} \cdot \frac{3}{4\pi} \cdot \frac{1}{3} + 2\pi \cdot \frac{1}{6} \cdot \frac{1}{4} \cdot \frac{3}{2\pi} \cdot \frac{2}{3} \\ + 2\pi \cdot \frac{1}{\sqrt{6}} \cdot \frac{\sqrt{3}}{4} \cdot \frac{1}{2\pi} \cdot \frac{1}{2} \cdot 2 \cos((E_2 - E_1)t) \int_0^\infty r^2 dr R_{10}(r) R_{21}(r)$$

We need one final integral:

$$\int_0^\infty r^2 dr \frac{2}{\sqrt{a_0^3}} e^{-r/a_0} \cdot \frac{1}{\sqrt{24} a_0^3} \frac{r}{a_0} e^{-r/2a_0}$$

$$= \frac{1}{\sqrt{6} a_0^4} \int_0^\infty dr r^3 e^{-3r/2a_0}$$

$$\int_0^\infty dr r^n e^{-kr} = \left(-\frac{d}{dk}\right)^n \int_0^\infty dr e^{-kr} = \left(-\frac{d}{dk}\right)^n \frac{1}{k} = \frac{n!}{k^{n+1}}$$

$$\frac{1}{\sqrt{6} a_0^4} \cdot \frac{3!}{\left(\frac{3}{2a_0}\right)^4} = \frac{\sqrt{6} \cdot 2^4}{3^4}$$

$$P(z > 0) = \frac{1}{4} + \frac{1}{6} + \frac{1}{12} + \cos[(E_2 - E_1)t]$$

$$\frac{1}{2} \left[ 1 + \frac{8}{27} \cos((E_2 - E_1)t) \right]$$

$$(c) P(r \rightarrow l_{nm}) = \int_{l_{nm}}^{\infty} r^2 dr \int d\Omega |\psi(\vec{r}, t)|^2$$

$\int d\Omega$  kills all cross terms with differing  $l, m$ .

↳ no phases for  $|\psi(\vec{r}, t)\rangle$  above

$$\begin{aligned} P(r \rightarrow l_{nm}) &= \int_{l_{nm}}^{\infty} r^2 dr \left[ \frac{1}{2} R_{10}^2(r) + \frac{1}{3} R_{21}^2(r) + \frac{1}{6} R_{22}^2(r) \right] \\ &= \frac{1}{2} \int_{l_{nm}}^{\infty} r^2 dr \left[ \frac{4}{a_0^3} e^{-2r/a_0} + \frac{1}{24a_0^4} r e^{-r/a_0} \right] \end{aligned}$$

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$$\int_{l_{nm}}^{\infty} dr r^n e^{-kr} = \left( -\frac{d}{dk} \right)^n \frac{e^{-kr}}{-k} \Big|_{l_{nm}}^{\infty} = \left( -\frac{d}{dk} \right)^n \frac{e^{-k(l_{nm})}}{k}$$

$$-\frac{d}{dk} \frac{e^{-kR}}{k} = \frac{e^{-kR}}{k^2} + \frac{R e^{-kR}}{k} = \left( \frac{1}{k^2} + \frac{R}{k} \right) e^{-kR}$$

$$\left( -\frac{d}{dk} \right)^2 \frac{e^{-kR}}{k} = \left( +\frac{1}{k^3} + \frac{R}{k^2} \right) e^{-kR} + \left( \frac{R}{k^2} + \frac{R^2}{k} \right) e^{-kR}$$

$$= \left( \frac{1}{k^3} + \frac{2R}{k^2} + \frac{R^2}{k} \right) e^{-kR}$$

$$\begin{aligned} \left( -\frac{d}{dk} \right)^3 \frac{e^{-kR}}{k} &= \left( \frac{1}{k^4} + \frac{4R}{k^3} + \frac{R^2}{k^2} \right) e^{-kR} + \left( \frac{R}{k^3} + \frac{2R^2}{k^2} + \frac{R^3}{k} \right) e^{-kR} \\ &= \left( \frac{1}{k^4} + \frac{5R}{k^3} + \frac{3R^2}{k^2} + \frac{R^3}{k} \right) e^{-kR} \end{aligned}$$

$$R = 1 \text{ nm}$$

$$P(r > 1 \text{ nm}) = \frac{2}{a_0^3} \left( \frac{1}{\left(\frac{2}{a_0}\right)^3} + \frac{2R}{\left(\frac{2}{a_0}\right)^2} + \frac{R^2}{\left(\frac{2}{a_0}\right)} \right) e^{-2R/a_0} \\ + \frac{1}{48a_0^4} \left( \frac{1}{\left(\frac{1}{a_0}\right)^4} + \frac{5R}{\left(\frac{1}{a_0}\right)^3} + \frac{3R^2}{\left(\frac{1}{a_0}\right)^2} + \frac{R^3}{\left(\frac{1}{a_0}\right)} \right) e^{-R/a_0}$$

We can simplify this a bit more:

$$P(r > 1 \text{ nm}) = \left( \frac{1}{4} + \frac{R}{a_0} + \frac{R^2}{a_0^2} \right) e^{-2R/a_0} \\ + \left( \frac{1}{48} + \frac{5R}{48a_0} + \frac{R^2}{16a_0^2} + \frac{R^3}{48a_0^3} \right) e^{-R/a_0}$$

$$\frac{R}{a_0} \approx \frac{10^{-9} \text{ m}}{0.5 \times 10^{-10} \text{ m}} \approx \frac{1}{5}$$

$$P(r > 1 \text{ nm}) \approx \left( \frac{1}{4} + \frac{1}{5} + \frac{1}{25} \right) e^{-2/5} + \left( \frac{1}{48} + \frac{1}{48} + \frac{1}{16 \cdot 25} + \frac{1}{(48)(125)} \right) e^{-1/5} \\ \approx \frac{1}{2} e^{-2/5} + \frac{1}{50} e^{-1/5} \approx \underline{\underline{0.35}}$$

$$(2) P(m_x) = \sum_{nl} |\langle nlm_x | \psi(t) \rangle|^2$$

We encounter  $\langle nlm_x | n'l'm' \rangle \propto \delta_{nn'} \delta_{ll'}$ ,

so most terms drop out.

For  $l=0$ ,  $\langle 00_x | 00 \rangle = 1$ ; only one state.

From table we can get  $\langle 10_x | 10 \rangle = 0$ ,  $\langle 10_x | 1, -1 \rangle = -\frac{1}{\sqrt{2}}$ ,

$\langle 11_x | 10 \rangle = \frac{1}{\sqrt{2}}$ ,  $\langle 11_x | 1, -1 \rangle = \frac{1}{2}$ ,  $\langle 1, -1_x | 10 \rangle = -\frac{1}{\sqrt{2}}$

and  $\langle 1, -1_x | 1, -1 \rangle = \frac{1}{2}$ .

$$P(m_x=1) = |\langle 211_x | \psi(t) \rangle|^2 = \left| \frac{1}{\sqrt{3}} e^{-iE_2 t} \langle 211_x | 210 \rangle + \frac{1}{\sqrt{6}} e^{-iE_2 t} \langle 211_x | 21, -1 \rangle \right|^2$$

$$= \left| \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{6}} \cdot \frac{1}{2} \right|^2 = \left| \frac{3}{2\sqrt{6}} \right|^2 = \frac{9}{24}$$

$P(m_x=1) = \frac{3}{8}$

$$\begin{aligned} P(m_x=-1) &= |\langle 21, -1_x | \psi(t) \rangle|^2 = \left| \frac{1}{\sqrt{3}} e^{-iE_2 t} \langle 21, -1_x | 210 \rangle + \frac{1}{\sqrt{6}} e^{-iE_2 t} \langle 21, -1_x | 21, -1 \rangle \right|^2 \\ &= \left| \frac{1}{\sqrt{3}} \cdot \left(-\frac{1}{\sqrt{2}}\right) + \frac{1}{\sqrt{6}} \left(\frac{1}{2}\right) \right|^2 = \left| \frac{1}{2\sqrt{6}} \right|^2 = \frac{1}{24} \end{aligned}$$

$$P(m_x=0) = |\langle 100_x | \psi(+)\rangle|^2 + |\langle 210_x | \psi(+)\rangle|^2$$

$$= \frac{1}{2} + \left| \frac{1}{\sqrt{3}} \langle 10_x | 10 \rangle + \frac{1}{\sqrt{6}} \langle 10_x | 1, -1 \rangle \right|^2$$

{ only  
l+m  
matter

$$= \frac{1}{2} + \left| \frac{1}{\sqrt{3}} \cdot 0 + \frac{1}{\sqrt{6}} \cdot \left(-\frac{1}{\sqrt{2}}\right) \right|^2 = \frac{1}{2} + \frac{1}{12}$$

$$P(m_x=0) = \frac{7}{12}$$

$$\text{Probs} \rightarrow \frac{3}{8} + \frac{1}{24} + \frac{7}{12} = 1 \quad \checkmark$$